

robotic rovers that could autonomously explore planet Mars. What NASA needed was an AI system that could navigate different environments using only images captured by a low-end camera. And this had to be achieved with limited computing, communications and power resources. Plus, the system would have to run on the single GPU chip that the rover had.

Not only did the team manage it, but now Versace's startup Neurala has an updated prototype of the AI system it developed for NASA, which can be applied for other purposes. The logic is that the same technology that was used by Mars rovers can be used by drones, self-driving cars and robots to recognise objects in their surroundings and make decisions accordingly.

Neurala too bets on deep learning as the future of its AI brain, but unlike most common solutions that run on online services backed by huge servers, Neurala's AI can operate on the computationally low-power chips found in smartphones.

In a recent press report, Versace hinted that their approach focuses on edge computing, which relies on onboard hardware, in contrast with other approaches that are based on centralised systems. The edge computing approach apparently gives them an edge over others. This is because the key to their system is hardware-agnostic software, which can run on several industry-standard processors including ARM, Nvidia and Intel.

Although their system has already been licensed and adapted by some customers for use in drones and cars, the company is very enthusiastic about its real potential in robot toys and household robots. They hope that their solution will ensure fast and smooth interaction between robots and users, something that Cloud systems cannot always guarantee.

Analogue intelligence, have you given it a thought

Shahin Farshchi, partner at investment firm Lux Capital, has a radically



Automated inspections by an intelligent drone with deep learning launched by Neurala, Aerialtronics and NVIDIA (Image courtesy: Neurala)

different view of AI and robots. He feels that all modern things need not necessarily be digital, and analogue has a great future in AI and robotics. In an article he wrote last year, he explained that some of the greatest systems were once powered by analogue, but it was abandoned for digital systems just because analogue was rigid and attempting to make it flexible made it more complex and reduced its reliability.

As Moore's law played its way into our lives, micro-electro-mechanical systems and micro-fabrication techniques became widespread, and the result is what we see all around us. He wrote, "In today's consumer electronics world, analogue is only used to interface with humans, capturing and producing sounds, images and other sensations. In larger systems, analogue is used to physically turn the wheels and steer rudders on machines that move us in our analogue world. But for most other electronic applications, engineers rush to dump signals into the digital domain whenever they can. The upshot is that the benefits of digital logic—cheap, fast, robust and flexible—have made engineers practically allergic to analogue processing. Now, however, after a long hiatus, Carver Mead's prediction of the return to analogue is starting to become a reality."

Farshchi claims that neuromorphic and analogue computing will make a comeback in the fields of

AI and robotics. Neural networks and deep learning algorithms that researchers are attempting to implement in robots are more suitable to analogue designs. Such analogue systems will make robots faster, smaller and less power-hungry. Analogue

circuits inspired by nature will enable robots to see, hear and learn better while consuming much less power.

He cites the examples of Stanford's Brains in Silicon project and University of Michigan's IC Lab, which are building tools to make it easier to build analogue neuromorphic systems. Some startups are also developing analogue systems as an alternative to running deep nets on standard digital circuits. Most of these designs are inspired by our brain, a noisy system that adapts according to the situation to produce the required output. This is in contrast to traditional hard-coded algorithms that go out of control if there is the slightest problem with the circuits running these.

Engineers have also been able to achieve energy savings of the order of 100 times by implementing deep nets in silicon using noisy analogue approaches. This will have a huge impact on the robots of the future, as they will not require external power and will not have to be connected to the Cloud to be smart. In short, the robots will be independent.

Training an army of robots using AI and exoskeleton suits

Kindred is a quiet but promising startup formed by Geordie Rose, one of the co-founders of D-Wave, a quantum computing company. According to an IEEE news report, Kindred is busy developing AI-driven robots that can possibly enable one human worker do the work of four.